

31 – Application Update and Distribution

How software reaches and stays current on up to 15,000 kiosk-locked till terminals across all 25 districts with no on-site IT, on poor links, without stopping trade: device provisioning, the update agent, signed releases, ring rollout, health gates, rollback, the version floor, and the back-office deployment that has to stay compatible with all of it.

Field	Value
Document	31 – Application Update and Distribution (design artefact; implements ADR-24; Stage 4 topic brought forward)
Version	v0.1 – draft for design review
Date	6 September 2026
Status	Draft. The management-agent product and the terminal platform (GMS-certified versus AOSP) are decision requests that must close together with the hardware selection in doc 14.
Inputs	ADR-24 (signed packages, ring deployment, install outside trading hours, server-enforced minimum version floor with grace); Q-10; Q-06, Q-07 (connectivity and power); T-07 (centrally specified hardware); doc 18 (device table), doc 19 (device identity, event model), doc 20 M6; Android Enterprise and Google developer-verification policy as published to August 2026.
Audience	Engineering lead, DevOps, the till track, helpdesk lead, hardware procurement.

Why this is a design problem and not an operations chore. A terminal in a shop with one cashier and no IT will never have an update installed by hand. Every update must arrive by itself, install itself when the shop is closed, prove it is healthy, and undo itself if it is not – and the central system must keep serving the versions that have not yet arrived. Get this wrong and the fleet fragments into versions nobody can support (Q-10).

1. What gets updated, and by which path

Component	Where it runs	Update path	User action
Web back office	Browser	Server deployment; the client reloads on version mismatch	None
Central services	Cloud or local hosting	Rolling deployment behind a load balancer; database migrations forward-only	None
Till application	Android terminal in the shop	This document: silent install by the device-owner management agent, ring rollout, health gate	None
Driver application	Android phone or terminal	Same agent and pipeline as the till	None
Management agent itself	Android terminal	Self-update via the same silent-install privilege; staged first	None
Terminal operating system and firmware	Android terminal	Vendor over-the-air channel, scheduled by the Federation through the vendor console; never automatic during trading	None
Reference data (catalogue, prices, rules, permissions, message catalogue)	Till local cache	Sync snapshots (doc 19 6.3), not this document	None

2. Constraints that shape the design

Constraint	Source	Consequence
Up to 15,000 terminals, no on-site IT, generally low computer familiarity	Q-05, S-02	Zero-touch: no prompts, no taps, no settings screens on the device
Poor and intermittent links (about 2 Mbps down at the weakest sites), power cuts	Q-06, Q-07	Resumable downloads, small packages, install only after full verification, never mid-transaction
24×7 trading capability, no scheduled downtime window	Q-04	Install outside each shop's own trading hours; never force a restart during a session
No indefinite deferral; minimum version floor with grace	ADR-24, Q-10	Central enforces a floor; the device cannot opt out
Hardware new and centrally specified; ARM/Android terminals	T-07, H-08	One managed platform; the agent can be a device owner from first boot
Google developer verification (2026–2027)	Section 3	Distribution must go through a device-owner management agent or Managed Google Play; direct APK sideloading is not a strategy
Single DevOps operating model	ADR-20	One console, one pipeline, automatic gates; no per-device handling

3. Android developer verification: what it means for a private till app

Google is introducing a requirement that applications installed on certified Android devices come from identity-verified developers, with enforcement beginning in September 2026 in a first group of countries and expanding globally in 2027. The published guidance, as of August 2026, is that enterprise deployment is exempt when the application is installed through an EMM (enterprise mobility management) device policy controller or published as a private application in Managed Google Play, while ordinary APK sideloading falls under the new friction. Devices without Google services (AOSP builds) are outside the scheme entirely.

Terminal platform	Verification exposure	Update path that works	Notes
GMS-certified Android terminal (Google services present)	In scope from 2027 for non-managed installs	Device-owner management agent (DPC) silent install, or Managed Google Play private app	Managed Play gives free hosting and delta updates but hands rollout timing to Google's update scheduler; the DPC path keeps timing under Federation control
AOSP terminal (no Google services)	Out of scope	Device-owner management agent silent install	Vendor-specific provisioning; fewer background services competing for memory; no Play protections

Decision: the design uses a device-owner management agent for silent install on either platform, so the platform choice can be made on hardware merits in doc 14. As a hedge costing almost nothing, the Federation registers as a verified developer and signs releases under that identity, so that even an unmanaged install path would remain available for recovery and development devices.

4. Architecture

Element	Role	Design
Provisioning	Make a new terminal a managed kiosk before it leaves central staging	Factory-reset terminal; QR-code or zero-touch enrolment sets the management agent as device owner; the agent installs the till app, applies the kiosk (lock-task) policy, disables unknown-source installs and developer options, and enrolls the device identity (doc 19 2.1). Staging is a table at the Federation, not a task at the shop.
Management agent (DPC)	The only thing on the device that can install software	Runs as device owner; polls the update service on the sync cadence and on a push wake-up when online; downloads, verifies, schedules and performs silent install; reports version and health; can lock or wipe a device reported stolen.
Update service (central)	Decide what each device should run	A small service owned by DevOps: release register, ring membership, rollout state machine, health gates, version floor, per-district status; reads the device table (doc 18) and writes device.app_version on report.
Release pipeline	Turn a build into a distributable release	Continuous integration builds the APK, runs tests, produces a software bill of materials, signs with the Federation release key held in a key-management service (never on a laptop), publishes to object storage behind the update service.
Till application	Cooperate with updates and enforce the floor	Reports version on every sync; refuses to sync (not to sell) below the floor after grace; runs forward-only local migrations on start; performs a self-test after any upgrade.
Fleet console	One screen for the DevOps engineer	Version distribution, rollout progress per ring and district, health gate status, stuck devices, halt and resume controls.

4.1 Choosing the management agent

Option	Fit	Concerns
Self-hosted open-source MDM used as device owner (Headwind MDM class)	Provider-neutral, runs on DigitalOcean or local hosting, silent install and kiosk on AOSP and GMS devices, one console	Community support; the Federation owns operations; verify the exact silent-install behaviour on the chosen terminal model before purchase
Terminal vendor's terminal-management system (for example the store and management consoles offered by POS terminal makers)	Tuned to the hardware; firmware and app updates in one place; silent install proven on that device	Vendor lock-in for the fleet; cloud outside the Federation's control; export of device data; cost per device
Commercial EMM (Android Enterprise partner)	Mature, supported, Managed Play integration	Per-device licensing at 15,000 devices; heavier than the single-DevOps model needs
In-house device-owner agent	Exactly the features needed and nothing else	The Federation builds and maintains a DPC; only sensible if the chosen open-source option fails validation

Recommendation (DR-1): validate the open-source device-owner MDM on two candidate terminal models during the Stage 3 spikes (silent install, lock task, wake on push, resumable download, wipe). Adopt it if it passes; fall back to the terminal vendor's system for firmware only, keeping the till app on the Federation-controlled path either way.

5. Release and rollout

5.1 Release discipline

Aspect	Design
Versioning	Semantic version for people; monotonically increasing Android versionCode for the platform. Android does not downgrade: a "rollback" is always a new, higher versionCode built from the previous good source.
Package size	Target under 30 MB; fonts and the message catalogue ship in the package (no first-run download); images are never in the package (they are catalogue data).
Compatibility	Every release supports the sync contract and snapshot formats of the two previous releases (N-2); central supports app versions from the floor to current; the API is backward compatible within that window.
Local schema	Migrations are forward-only, idempotent, run at start before the UI; the agent triggers a local database backup to device storage before install; a failed migration reports and the health gate catches it.

Aspect	Design
Signing	One release key in the key-management service; the agent and the till verify signature and package hash before install; unsigned or mis-signed packages are rejected and reported.

5.2 Rings

Ring	Membership	Purpose	Exit gate
0 Lab	Federation test bench: every supported terminal model, simulator, one printer of each type	Functional and peripheral regression	All automated tests pass; manual checklist signed
1 Canary	About 1% of shops, chosen to cover every district, every size band, both connectivity profiles	Real-world health under load	48 hours with health metrics inside thresholds
2 Wave	One rollout wave (a group of districts) at a time	Controlled spread	24 hours per wave inside thresholds
3 Fleet	Everyone else	Completion	Fleet at target version above 97% within the grace period; stragglers to the helpdesk queue

5.3 Health gates

Metric	Source	Threshold (initial)	Action on breach
Crash-free session rate	Till telemetry on sync	≥ 99.5% on the new version	Halt rollout; alert DevOps
Sync success rate	Update service and ingest	No worse than the previous version by more than 0.5 points	Halt
Receipt issuance rate per till-hour	Ingest projections	No drop beyond normal variance versus the same shops last week	Halt; investigate
Peripheral self-test failures	Till telemetry	≤ 0.5% of updated devices	Halt
Devices stuck in install or migration	Agent reports	≤ 0.2%	Automatic re-install of previous version on those devices; helpdesk ticket

5.4 On-device sequence

Step	Action	Safeguard
1	Agent learns a release applies to this device's ring	Poll on sync cadence; push wake-up when online
2	Download in the background, resumable, throttled below 50% of measured bandwidth	Never competes with sync uploads; pauses on metered or failing links
3	Verify signature and hash	Reject and report on mismatch
4	Wait for the install window: outside the shop's trading hours (location.trading_hours), no open till session, battery above 40% or on mains	Window is per shop; a shop with 24-hour trading uses its configured quiet hour
5	Back up the local database; install silently; restart the till app in lock-task mode	Install is atomic on Android; the backup covers a failed migration
6	Self-test: open database, run migrations, print a test line if a printer is present, scanner check, sync handshake	Failure reported immediately
7	Report version and health on next sync	Feeds the health gate and fleet console
8	On a failed self-test or crash loop: agent installs the previous good release (higher versionCode rebuild) and raises an ALERT	Shop reopens on a working till; helpdesk sees the device

6. The version floor

ADR-24 forbids indefinite deferral. The floor is what makes that true without stopping trade.

Aspect	Design
Definition	The lowest app version central will accept for sync. Set per release by DevOps; normally N-2.
Grace	When a release raises the floor, devices below it get a configurable grace period (default 14 days) during which sync continues and the till shows a persistent, non-blocking notice to the supervisor.
After grace	Sync is refused with a clear on-screen reason; selling continues offline (blocking sales would punish the shop for a fleet problem). The device appears in the helpdesk queue automatically. The helpdesk boundary of Q-09 (no connectivity troubleshooting) still applies; the fix is almost always connectivity, and the agent installs as soon as the device reaches central.
Escalation	A device below the floor for more than the grace period plus 7 days is escalated to the MPCS administrator and the district onboarding team, since a shop that cannot sync for three weeks has a bigger problem than its app version.
What the floor never does	It never wipes, locks or disables a till; it never applies to the driver app's proof-of-delivery capture; it never triggers during the pilot go/no-go window without an explicit decision.

7. Central and web deployment

Aspect	Design
Rolling deployment	Two or more instances behind the load balancer; deploy one at a time; health check before traffic; database migrations applied before code that needs them, written to be compatible with the previous code version (expand-then-contract).
Web client	Served with content hashes; the shell compares its build version with the API's advertised version and prompts a reload on mismatch; API responses stay compatible with the previous client for at least one deployment.
Sync ingest compatibility	Central accepts the envelope and snapshot formats of every app version above the floor; format changes are additive within the N-2 window.
Feature exposure	New behaviour that changes what a shop sees is switched on per ring through configuration, not by deploying different code to different shops.
Local hosting variant	The same pipeline; the update service and package store run on the local host; nothing in the design depends on a cloud-only service.

8. Security and fleet operations

Concern	Design
Release key custody	Key-management service with dual control for signing; rotation plan with a transition period where both keys are trusted.
Agent authority	Agent authenticates with the device credential (doc 19); update instructions are signed by central; the agent will not act on unsigned instructions even from a trusted network.
Kiosk integrity	Lock-task allow-list contains the till app, the agent and the vendor's peripheral services only; USB debugging, unknown sources and factory reset are policy-disabled; safe-boot disabled.
Theft and loss	Remote lock and wipe; local database encrypted with a keystore-held key so a stolen terminal yields no data.
Fleet visibility	Fleet console reads device.app_version, last_seen_at, ring, health; per-district and per-wave views; stuck-device queue integrated with the helpdesk.
Onboarding volume	About 40 shops a week (S-07): provisioning is a staging-table process with QR enrolment and a checklist; the terminal reaches the shop already managed, enrolled and on the current release.
Firmware	Vendor OTA scheduled through the vendor console in the same quiet-hour windows, one wave at a time, never on the same day as an app rollout.

9. Decision requests

#	Point	Proposed	Decider	Needed by
DR-1	Management agent product	Open-source device-owner MDM validated in Stage 3 spikes; vendor console for firmware only	Engineering lead, DevOps	Hardware selection (doc 14)

#	Point	Proposed	Decider	Needed by
DR-2	Terminal platform: GMS-certified or AOSP	Decide on hardware merits; the update path works on both	Hardware procurement	Doc 14
DR-3	Register the Federation as a verified Android developer	Yes, now; low cost hedge	Federation admin	Before pilot
DR-4	Ring sizes, gate thresholds, grace period	1% canary, wave by district, 14-day grace, thresholds of 5.3	DevOps with Federation	Before pilot
DR-5	Quiet-hour policy for 24-hour shops	Configured per location; default 02:00–04:00	MPCS via configuration register	Onboarding

End of document. Draft for design review; Stage 4 operational design refines the fleet console and helpdesk runbooks against this design.